Matemàtiques B

4t ESO

SOLUCIONS DE LA FITXA · UNITAT 3

Equacions i inequacions

FentMates

Intenta resoldre cada exercici abans de mirar-ne la solució. Pots comprovar les equacions i les inequacions amb les eines interactives dels apunts de la unitat.

1 Equacions de segon grau

1. ★☆☆

a) $\boxed{x = \pm 4}$ **b)** $\boxed{x = 0,\ x = -4}$ **c)** $\boxed{x = \pm 5}$

d) No té solució. **e)** $\boxed{x = 0,\ x = \dfrac{3}{5}}$ **f)** $\boxed{x = \pm\dfrac{1}{2}}$

2. ★☆☆

a) $\boxed{x = 2,\ x = 3}$ **b)** $\boxed{x = 3,\ x = -5}$ **c)** $\boxed{x = 3,\ x = \dfrac{1}{2}}$

d) $\boxed{x = 3}$ (doble) **e)** No té solució ($\Delta < 0$). **f)** $\boxed{x = \dfrac{1}{2},\ x = -\dfrac{1}{3}}$

3. ★★☆

a) $x^2 + x - 6 = 0$: $\boxed{x = 2,\ x = -3}$ **b)** $x^2 - 5x - 14 = 0$: $\boxed{x = 7,\ x = -2}$

c) $x^2 + 5x - 6 = 0$: $\boxed{x = 1,\ x = -6}$ **d)** $2x^2 - 3x - 9 = 0$: $\boxed{x = 3,\ x = -\dfrac{3}{2}}$

4. ★★☆

a) $\Delta = -56$: $\boxed{\text{cap}}$ **b)** $\Delta = 0$: $\boxed{\text{una (doble)}}$ **c)** $\Delta = 5$: $\boxed{\text{dues}}$

$\Delta = 36 - 4k = 0 \Rightarrow \boxed{k = 9}$.

5. ★★★ $(x - 3)(x + 5) = 0 \Rightarrow \boxed{x^2 + 2x - 15 = 0}$ (o qualsevol múltiple).

2 Equacions biquadrades

6. ★☆☆

a) $t = 9,\ t = 1$: $\boxed{x = \pm 3,\ x = \pm 1}$ **b)** $t = 9,\ t = -4$: $\boxed{x = \pm 3}$

c) $t = -1,\ t = -2$: no té solució. **d)** $t = 4,\ t = \frac{1}{4}$: $\boxed{x = \pm 2,\ x = \pm\dfrac{1}{2}}$

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7. ★★☆

a) $x^2(x^2-8)=0$: $\boxed{x=0,\ x=\pm 2\sqrt{2}}$

b) $x^4+x^2-12=0$: $\boxed{x=\pm\sqrt{3}}$

c) $\boxed{x=\pm 2}$

3 Equacions de grau major que 2

8. ★★☆

a) $\boxed{x=1,\ x=2,\ x=3}$

b) $\boxed{x=1,\ x=-1,\ x=-2}$

c) $(x-1)^2(x+2)=0$: $\boxed{x=1,\ x=-2}$

d) $\boxed{x=2,\ x=-\frac{1}{2},\ x=-2}$

9. ★★☆

a) $(x-1)(x+1)(x+2)(x-3)=0$: $\boxed{x=\pm 1,\ x=-2,\ x=3}$

b) $x(x+2)(x-2)=0$: $\boxed{x=0,\ x=\pm 2}$

c) $x(x+1)^2(x-1)=0$: $\boxed{x=0,\ x=-1,\ x=1}$

4 Equacions factoritzades

10. ★☆☆

a) $\boxed{x=4,\ x=-1}$

b) $\boxed{x=0,\ x=-\frac{3}{2},\ x=5}$

c) $\boxed{x=1,\ x=-7}$

d) $\boxed{x=-2,\ x=\frac{1}{4}}$

11. ★★☆

a) $\boxed{x=\pm 4,\ x=-3}$

b) $\boxed{x=1,\ x=2}$ ($x^2+5=0$ no en dona)

c) $\boxed{x=0,\ x=4,\ x=-2}$

d) No és «$=0$»: $x^2-x-6=0$, $\boxed{x=3,\ x=-2}$

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5 Equacions amb fraccions algebraiques

12. ★☆☆

a) $\boxed{x = 1}$

b) $\boxed{x = 3}$

c) $x^2 - 2x - 1 = 0$: $\boxed{x = 1 \pm \sqrt{2}}$

d) $x^2 - 5x + 4 = 0$: $\boxed{x = 1,\ x = 4}$

13. ★★☆

a) $x^2 - 3x - 10 = 0 \Rightarrow x = 5,\ x = -2$; $x = -2$ anul·la un denominador: $\boxed{x = 5}$

b) $x + 5 = 6 \Rightarrow x = 1$, que anul·la $x - 1$: $\boxed{\text{no té solució}}$

14. ★★★ m.c.m. $= 6x(x+1)$: $6(x+1)^2 - 6x^2 = 5x(x+1) \Rightarrow 5x^2 - 7x - 6 = 0 \Rightarrow \boxed{x = 2,\ x = -\frac{3}{5}}$ (totes dues valen).

6 Equacions amb radicals

15. ★☆☆

a) $\boxed{x = 4}$

b) $\boxed{x = 13}$

c) $\boxed{x = \pm 4}$

d) No en té: una arrel no és negativa.

16. ★★☆

a) $x^2 - x - 2 = 0 \Rightarrow x = 2,\ x = -1$; $x = -1$ no val ($\sqrt{1} \neq -1$): $\boxed{x = 2}$

b) $x^2 - 15x + 50 = 0 \Rightarrow x = 5,\ x = 10$; $x = 10$ no val ($10 + 3 \neq 7$): $\boxed{x = 5}$

c) $x^2 - 3x - 4 = 0 \Rightarrow x = 4,\ x = -1$; $x = -1$ no val ($1 + 1 \neq 0$): $\boxed{x = 4}$

d) $x^2 - 10x + 9 = 0 \Rightarrow x = 1,\ x = 9$; $x = 1$ no val ($2 \neq -2$): $\boxed{x = 9}$

17. ★★★

a) $x + 7 = 1 + 2\sqrt{x} + x \Rightarrow \sqrt{x} = 3$: $\boxed{x = 9}$

b) $x - 12 = -8\sqrt{x - 3} \Rightarrow x^2 - 88x + 336 = 0 \Rightarrow x = 4,\ x = 84$; $x = 84$ no val ($13 + 9 \neq 4$): $\boxed{x = 4}$

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7 Equacions exponencials

18. ★☆☆

a) $\boxed{x = 5}$ **b)** $\boxed{x = 5}$ **c)** $\boxed{x = -1}$

d) $\boxed{x = \frac{3}{2}}$ **e)** $\boxed{x = \pm 2}$ **f)** $\boxed{x = -4}$

19. ★☆☆

a) $\boxed{x = \frac{\log 10}{\log 2} \approx 3{,}322}$ **b)** $\boxed{x = \frac{\log 50}{\log 3} \approx 3{,}561}$

c) $\boxed{x = 1 + \frac{\log 30}{\log 7} \approx 2{,}748}$ **d)** $\boxed{x = \frac{\log 2}{\log 1{,}05} \approx 14{,}207}$

20. ★★☆

a) $3 \cdot 2^x = 24$: $\boxed{x = 3}$ **b)** $8 \cdot 3^x = 72$: $\boxed{x = 2}$ **c)** $\frac{6}{5} \cdot 5^x = 30$: $\boxed{x = 2}$

21. ★★★

a) $t = 2^x$: $t = 1$, $t = 4$; $\boxed{x = 0,\ x = 2}$

b) $t = 3^x$: $t = 1$, $t = 9$; $\boxed{x = 0,\ x = 2}$

c) $t = 4$ ($t = -3$ no val): $\boxed{x = 2}$

8 Equacions logarítmiques

22. ★☆☆

a) $\boxed{x = 1000}$ **b)** $\boxed{x = 15}$ **c)** $\boxed{x = 1}$ **d)** $\boxed{x = \frac{26}{3}}$

23. ★★☆

a) $\boxed{x = 5}$ **b)** $\boxed{x = 3}$

c) $x^2 - 5x + 6 = 0$: $\boxed{x = 2,\ x = 3}$ **d)** $x^2 + 9x - 10 = 0$; $x = -10$ no val: $\boxed{x = 1}$

24. ★★★

a) $x + 6 = 10(x - 3)$: $\boxed{x = 4}$ **b)** $x^2 - 2x - 8 = 0$; $x = -2$ no val: $\boxed{x = 4}$

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9 Inequacions de primer grau

25. ★☆☆

a) $\boxed{(4, +\infty)}$



b) $\boxed{[-2, +\infty)}$



c) $\boxed{(-\infty, -4)}$



d) $\boxed{(-\infty, -4]}$



26. ★★☆

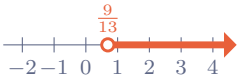
a) $0 \geq 0$, sempre: $\boxed{\mathbb{R}}$



b) $\boxed{(-\infty, 3)}$



c) $\boxed{(\frac{9}{13}, +\infty)}$



d) $-1 > 5$, mai: $\boxed{\varnothing}$

27. ★★☆ $15 + 0{,}5x > 40 \Rightarrow x > 50$: amb $\boxed{\text{més de 50 GB}}$ és més cara la B.

10 Inequacions de segon grau o superior

28. ★★☆

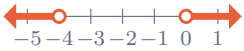
a) $\boxed{(-3, 3)}$



b) $\boxed{(-\infty, -2] \cup [5, +\infty)}$



c) $\boxed{(-\infty, -4) \cup (0, +\infty)}$



d) $\boxed{(-1, 3)}$



e) Només $x = 2$: $\boxed{\{2\}}$



f) $\Delta < 0$ i $a > 0$, sempre positiu: $\boxed{\mathbb{R}}$



29. ★★★

a) $\boxed{(-1, 0) \cup (1, +\infty)}$



b) $\boxed{(-\infty, -2] \cup [1, 4]}$



c) $x(x+1)(x-3) \geq 0$: $\boxed{[-1, 0] \cup [3, +\infty)}$



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30. ★★☆ $x(x+3) < 40 \Rightarrow x^2+3x-40 < 0 \Rightarrow x \in (-8,5)$. Com que $x > 0$: $\boxed{x \in (0,5)}$.

11 Problemes

31. ★★☆ $x(x+1) = 156 \Rightarrow x^2+x-156 = 0 \Rightarrow x = 12$ ($x = -13$ no és natural). Són $\boxed{12 \text{ i } 13}$.

32. ★★☆ $x + y = 17$ i $x^2+y^2 = 169$: $x^2+(17-x)^2 = 169 \Rightarrow x^2-17x+60 = 0 \Rightarrow$ $\boxed{12 \text{ cm i } 5 \text{ cm}}$.

33. ★★☆ $5000 \cdot 1{,}03^t = 6000 \Rightarrow t = \dfrac{\log 1{,}2}{\log 1{,}03} \approx 6{,}17$: calen $\boxed{7 \text{ anys}}$.

34. ★★★ $2^{t/3} = 32 = 2^5 \Rightarrow \frac{t}{3} = 5 \Rightarrow$ $\boxed{t = 15 \text{ hores}}$.

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